Chu Kai Le

chukaile.work@gmail.com | linkedin.com/in/ckail | kaishx.github.io

Education

National University of Singapore (NUS) Aug 2026 – May 2029
B.Eng. Mechanical Engineering, Second Major in Computing, Minor in Mathematics *Singapore*

- **Awards:** CDE Engineering Scholars Programme (Full-ride, non-bonded Scholarship)
- **Relevant Coursework:** CS2040DE Data Structures and Algorithms

NUS High School of Mathematics and Science Jan 2018 – Nov 2023
NUS High Diploma (High Distinction) *Singapore*

- **Major CAP:** Mathematics: 4.9/5.0, Chemistry: 4.9/5.0, Physics: 4.6/5.0

Experience

Genk Capital Feb 2026 – May 2026
Quantitative Research Intern *Singapore*

- Derived a microstructure signal for price-level collapses in large-tick assets; signal delivered a consistent **+19pp lift** in collapse persistence across all flow regimes (**40%** → **59%** in the tradable regime), validated over **106 days** of L2 LOB data.
- Modeled passive spread-capture entries using a **three-outcome framework** (fill, adverse selection, or flat); demonstrated positive net EV across all queue positions after accounting for one-way taker fees.
- Conducted **survival analysis** on **605K L2 updates** to validate a **97ms median time-to-collapse**, confirming the signal was operationally viable for real-time execution.
- Ran parameter grid sweeps across regimes and instruments; identified regime-dependent fill probability as the primary EV sensitivity driver and systematically excluded suboptimal signal configurations.

Projects/Competitions

Walk-Forward Analysis Engine | *Python, NumPy, Numba*

- Built a walk-forward pairs-trading research framework incorporating rolling in-sample/out-of-sample validation, Kalman-filter hedge ratios, stationarity testing, and transaction-cost modelling.
- Evaluated strategy robustness through statistical filters, parameter sensitivity analysis, and performance comparisons across research configurations.

Market Microstructure Research & Trading Systems | *Python*

- Screened perpetual instruments across **15+ venues** using LOB metrics (spread, OFI, lead-lag) to assess market-making viability and adverse-selection risk.
- Developed and backtested market-making strategies using parameter robustness checks, single-trade dominance tests, and P&L timescale decomposition to assess overfitting and strategy viability.
- Engineered a multi-venue real-time market data pipeline covering **330+ instruments**; integrated reconciliation, state tracking, staleness guards, and circuit breakers for live strategy testing.

IMC Prosperity 04 (2026) | *Python*

- Placed **123rd / 18,803** globally in university algorithmic trading competition; led strategy for a 5-person team of IMO/IOI/IChO gold medalists.

Technical Skills/Awards

Programming: Python (Pandas, NumPy, Numba), SQL, VBA, Bash
Systems & Tools: Linux, Git, SSH, real-time data pipelines
Quantitative Methods: Statistical modelling, survival analysis, backtesting, time-series analysis
Awards: Singapore Armed Forces National Serviceman of the Year (2026)
Interests: Valorant (Peak Top 0.8%), Trackmania (Peak Top 1 SG, Top 1% Asia), Bouldering (V7), Motorcycling